RESEARCH NOTE R0.70J · EXACT ALGEBRAIC GATE FOR THE DEVIATORIC DIAGONAL CORRELATION

Trace freedom removes only the scalar part; Helicity does not cancel the deviatoric diagonal either

R0.70I left one minimal question: does the exterior trace-free strain become automatically orthogonal to the core high-frequency vorticity covariance because of incompressibility, pure helicity, or angular averaging? This release computes the complete symbol and gives a positive-correlation witness. The answer is no; the next step must use source evolution from the actual equation or quantitative anisotropy information.

Status · R0.70J exact certificate, independent internal PASS review, and formal figure complete

Universal algebraic-null branch closed

Version v0.70J · 2026-08-24
Exact checks: all 5 groups passed
Focused regression: 14 / 14
Figure validation: 22 checks passed
Public framing: an internal research report, not a theorem chapter

00 / EXACT GATE RESULT

The deviatoric projection provides no second cancellation

For any symmetric trace-free matrix S and vector w, define

$$\mathring{Q}(w) = w \otimes w - \frac{|w|^2}{3} I.$$

The exact contraction is

$$\boxed{S : \mathring{Q}(w) = w^\top S w.}$$

Trace freedom gives only $S : I = 0$; it does not make two objects in the same five-dimensional symmetric trace-free tensor space orthogonal. If $S \neq 0$, choosing w along a positive eigendirection gives a strictly positive value.

Route decision. There is no universal algebraic null structure based only on trace freedom, incompressibility, fixed helicity, phase or angular averaging, and a physical cutoff. This conclusion does not depend on a denser Fourier grid or additional helicity labels.

01 / ROUTE FROM R0.70A TO R0.70I

Nine consecutive gates reduce the problem to one deviatoric diagonal term

Stage	Exact information retained	What is closed or still missing
R0.70A	A moving scale label leaves no $\dot{r}$ boundary term in the total vorticity-stretching identity.	Only a non-explicit robust neighborhood is obtained around scale ratio four.
R0.70B	Only a forward estimate connects the signed physical annuli to the nonnegative filtered reservoirs.	The kinematic reverse estimate and the tested translation-invariant quadratic normal form both fail.
R0.70C	A genuine smooth small-data Navier–Stokes evolution can carry the initial sign into a short time interval.	The signed integral can vanish while the absolute activity remains strictly positive; spatial-inversion parity and the linear heat layer do not repair this defect.
R0.70D	All cover averages are positive at the fixed scale.	This still does not control the unresolved negative mass; the witness at this stage is an abstract scalar density.
R0.70E	The attribution of objects in Yu's paper is corrected, and the paper's remainder work and the project's moving-shell contraction are checked separately.	Both admit smooth small-data witnesses for which the signed quantity is zero and the absolute quantity is positive.
R0.70F	The affine Taylor expansion of the fixed exterior source does yield a scale gain.	Unweighted packing after taking absolute values term by term still accumulates linearly; the construction lies only on the initial face.
R0.70G	The square estimate for adjacent exterior-source coefficients holds.	The critical dilation defect is transferred to the dual core-moment estimate.
R0.70H	At a fixed time, the core moments have an unweighted scale-variation bound.	The parabolic coordinate adds an amplification by r_k^{-2}, and taking absolute values in the moment evolution encounters the stretching term again.
R0.70I	The temporal gap is reduced to the $s^{-1/2}$ Hardy weight; the frozen-low mixed term closes at the Leray energy level.	The moving-low accumulation and the deviatoric high–high diagonal remain.

These intermediate results are route decisions, not independent regularity theorems. R0.70J addresses only the final precise question: whether the deviatoric diagonal has a deeper algebraic cancellation from helicity.

02 / TENSOR AND CUTOFF SYMBOL

A physical cutoff retains the zero-output term from conjugate frequencies

Incompressibility only requires the vorticity amplitude at one frequency to be perpendicular to the wave vector. Unless the additional exceptional compatibility condition $P_\xi S P_\xi = 0$ holds, the plane $\xi^\perp$ still contains directions on which the quadratic form is nonzero.

For a real field W, a constant source S, and a physical cutoff χ, the exact Fourier form is

$$\int \chi W^\top S W \, dx = \frac{1}{(2\pi)^6} \iint \widehat{\chi}(-\xi-\eta)\widehat{W}(\xi)^\top S \widehat{W}(\eta) \, d\xi d\eta.$$

Multiplication by χ is physical-space localization, not a Fourier projection. A real wave contains both k and $-k$, so the diagonal covariance at $k+(-k)=0$ does not disappear under second-harmonic averaging.

03 / PURE-HELICITY AND ANGULAR AUDIT

Both helicity signs give the same phase-averaged kernel

For a real pure-helicity wave with direction ξ and helicity $\sigma \in \{\pm 1\}$, the normalized phase average is

$$\boxed{K_S(\xi) = -\xi^\top S \xi.}$$

This kernel is independent of σ and is even under $\xi \mapsto -\xi$. Opposite frequencies and homochiral projection cannot cancel it.

take

$$S_0 = \mathrm{diag}(1/2, 1/2, -1), \qquad \omega_\sigma = \sqrt{2}(e_1 \cos\theta - \sigma e_2 \sin\theta),$$

Direct calculation gives $S_0 : \mathring{Q}(\omega_\sigma) \equiv 1$. The signed mean over the full sphere is indeed zero, but the positive-part mean remains

$$\langle (K_{S_0})_+ \rangle_{S^2} = \frac{\sqrt{3}}{9} > 0.$$

Therefore, taking the signed average before imposing isotropy and summing positive parts scale by scale are two different problems.

04 / COMPACT PHYSICAL REALIZATION

The exterior strain and core vorticity can be separated exactly in physical space

The report uses compactly supported homotopy localization to construct a smooth divergence-free exterior vorticity source whose induced velocity produces the constant strain S_0 throughout the open core; another compactly supported divergence-free carrier has the prescribed Beltrami-type vorticity in the core. All return vorticity is retained, and the source and core supports are strictly separated.

For every nonzero nonnegative core cutoff, the local contraction remains strictly positive. A compactly supported carrier and a strictly band-limited object cannot be the same function; this is an unavoidable uncertainty-principle boundary. A strict Littlewood–Paley annular realization still requires a separate high-frequency pseudolocal error estimate.

The same construction can serve as smooth compactly supported initial data. By continuity, the small-amplitude Navier–Stokes solution retains the positive sign near the initial face, but the resulting time window shrinks like r^2 under scaling.

05 / CRITICAL SCALE LEDGER

The source–core Cauchy product is already sharp in the critical coordinates

At outer scale $R = \Lambda r$, amplitude a, and a parabolic time window of length r^2, the normalized source coefficient and core moment give

$$\int r^{-1}|c_r|^2 dt \sim a^2 r\Lambda^{-4}, \qquad \int r|\mathcal{N}_r|^2 dt \sim a^4 r^{-1},$$

$$\int c_r : \mathcal{N}_r \, dt \sim a^3\Lambda^{-2} > 0.$$

The two squared norms scale with opposite powers of r, so their product gains nothing. This shows that ordinary Cauchy duality is already sharp at the critical scale; more absolute-value estimates of the same kind do not automatically produce the missing small factor.

06 / JOURNAL FIGURE

One figure shows the helicity witness, the angular positive part, and critical sharpness together

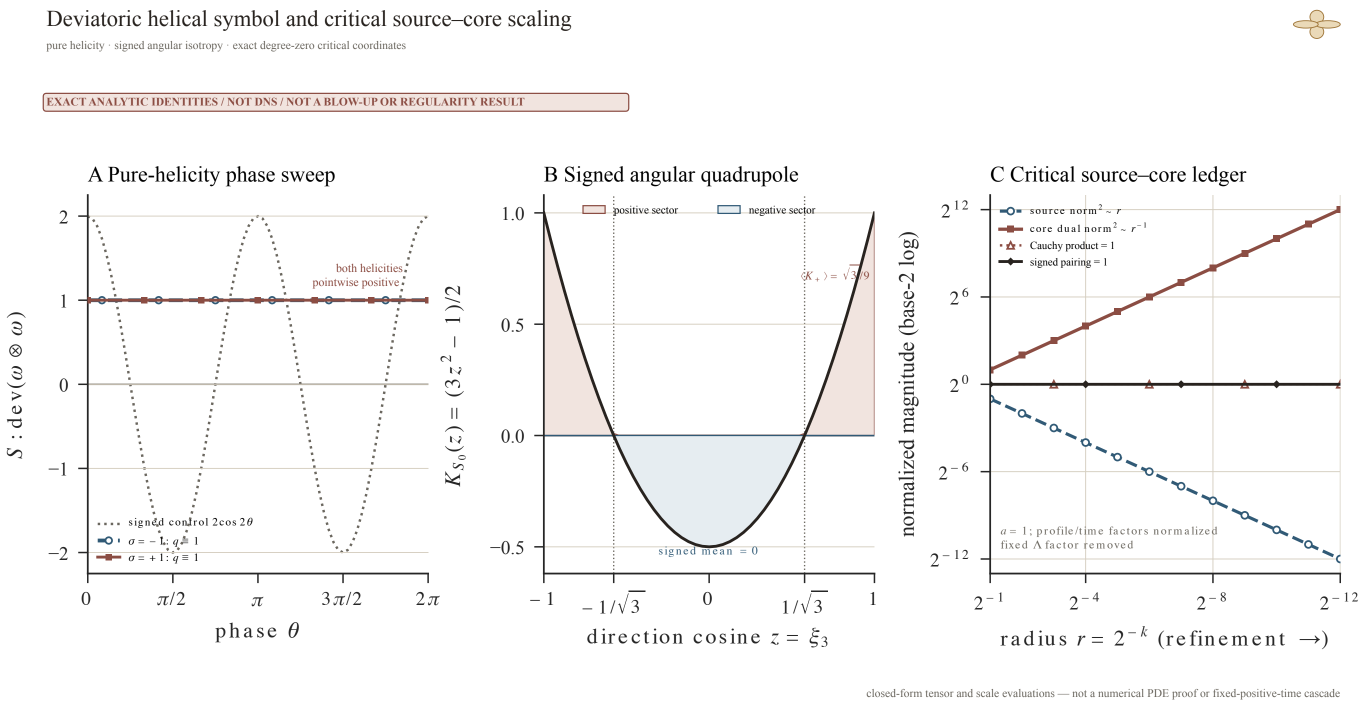

Figure R0.70J. A: The positive-correlation witnesses for both helicities equal one at every phase. B: The signed mean over the full sphere is zero, whereas the positive-part mean is $\sqrt{3}/9$. C: The source squared norm and core dual squared norm scale with opposite critical exponents, while the Cauchy product and signed pairing remain constant.

Download the vector PDF figure · Download the 600 dpi PNG

07 / RESEARCH VALUE AND LITERATURE BOUNDARY

The value is to turn a vague ‘perhaps there is a null structure’ into a falsifiable proposition

The bounded primary-literature audit covers harmonic tensors, helical triads, homochiral cascades, vorticity-direction coherence, strain–vorticity alignment, the pressure Hessian, and localized regularity criteria. The existing results provide representation theory, conditional regularity, or statistical geometry; among the ten primary sources audited here, I found no universal orthogonality theorem of the required kind. This is a bounded search finding, not a theorem that no such result exists anywhere in the literature, and it makes no priority claim.

The contribution of R0.70J is not that the major problem has advanced by a large step, but that it closes a mechanism that could otherwise misdirect later work: once the minimal Beltrami witness is rigorously nonzero, there is no reason to keep enlarging the angular sample or the set of helicity labels in search of the same cancellation.

Equation-specific compensation remains worth studying: how the true low-frequency source evolves with pressure and subgrid terms, and whether summing adjacent scales produces cancellations invisible in individual terms.

08 / CLAIM BOUNDARY

The algebraic branch is closed, not the Millennium problem

PROVED **Exact deviatoric and helicity symbols** All five groups of SymPy checks, the independent audit, and the hash manifest passed.	PROVED **Compactly supported positive-correlation realization** The return field and the source–core support separation are recorded in full.
OPEN **Fixed positive terminal time** Small-data continuity retains only a shortened initial time window.	OPEN **Leray-to-critical closure** The strict annular realization, pressure trajectory, and weak-solution limit are all unfinished.

This release does not construct a singularity, prove global smoothness, or produce a cascade along one finite-energy solution. It cannot be called a partial solution of the Millennium problem.

09 / REPRODUCIBILITY AND SOURCES

The report, literature audit, independent review, certificate, and figure package are archived together

View the complete mathematical report · View the bounded primary-literature audit · View the independent internal review

View the exact certificate and SHA-256 manifest · View the ten-file journal figure package · 下载同步 PDF (Chinese PDF) · View the Clay Mathematics Institute's official problem statement

The certificate producer covers only finite exact algebra. Smooth support, strict band-limiting error, small-data theory, and literature completeness are stated separately in the claim boundary and all remain outside the scope of the computer proof.